

IOT BASED SYSTEMS

UEC 640

**IOT-BASED VEHICLE SPEED DETECTION AND ALERT SYSTEM
FOR CAMPUSES**

Submitted By:

Natasha 102306129

Deeksha Sharma 102306125

Submitted To:

Dr. Amanpreet Kaur



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING

Thapar Institute of Engineering & Technology, Patiala

Session: Jan to May, 2026

Abstract

Speeding remains a major contributor to road traffic accidents, especially at vulnerable locations such as schools, hospitals and busy city streets. It's evident that maintaining safe vehicle speeds at such areas is essential, but the current available systems are either costly to roll out or lack features. The project helps to solve this problem by developing an IoT-based vehicle speed monitoring and alert system particularly for campus that tracks speed in real time and provides immediate alerts in case of speed infringement. The system uses a pair of ultrasonic sensors that are fixed at two points along the road. As vehicles drive through the two sensors, it calculates the time of travel and hence the speed. This data is processed on an ESP32 micro-controller and compared with a pre-determined threshold. In case the vehicle is over-speeding, an alarm will be triggered on a display board and an E-mail will be sent to the concerned authority. Not only can it detect, it also has a GPS to record the location of the offence and a camera to record the vehicle for evidence. This thus makes it not only a speed detector but also an enforcement tool. The system was tested to be able to detect the speed and react in a timely fashion. The components used are cheap and easy to install, meaning this solution could well be used at several points of interest along a roadside. In all, the project provides a practical, affordable solution to enhance road safety and assist authorities in tracking down violations within the campus.

Keywords: - ESP32, IoT, WiFi Communication, Ultrasonic Sensors, SMTP, IP Webcam, etc.

Introduction

The number of vehicles on roads has grown dramatically over the past decade, and with that, road safety has become a serious concern. Of all the factors behind road accidents, over-speeding stands out as one of the most dangerous. When a driver is going too fast, reaction time drops and the impact of any collision becomes far more severe. This risk is especially high near schools, hospitals, and busy residential areas where pedestrians are frequently crossing.

For years, speed monitoring was largely done through radar guns and manual checks by traffic personnel. Radar systems do work well in terms of accuracy, but they are expensive and difficult to scale across many locations. This has pushed researchers to look at IoT and embedded systems as more practical alternatives. For example, an early approach used a GSM module to notify the vehicle owner whenever a speed limit was crossed. It was simple and functional, but it only sent alerts to the owner — there was no way to track where the violation happened or share the data with authorities.

Another method used infrared sensors to time a vehicle passing between two points, triggering a buzzer when the speed was too high. While it worked for the basic task, it was purely local — no remote notifications, no logging, and no tracking. More capable system brought GPS and cloud connectivity into the picture, allowing real-time monitoring from a central location. The downside was its dependence on a stable internet connection, which isn't always available on all roads, especially in semi-urban or rural areas. Some research has even combined speed detection with accident detection mechanisms, though these tend to focus on what's happening inside the vehicle rather than monitoring from the roadside.

Looking at these existing solutions together, a pattern becomes clear: most systems do one or two things well but leave gaps elsewhere. Some lack communication features, while some are just built for highways or roads, others lack evidence collection, and many fall short when internet access isn't reliable.

That's what this project sets out to fix. The proposed system uses ultrasonic sensors and an ESP32 microcontroller to detect speed at the roadside of the university campus — no modifications to the vehicle needed. If a violation is detected, it sends an E-Mail notification to the respective authority of the campus, logs the GPS coordinates of the incident, and takes a photo of the vehicle as evidence. The goal was to combine all of these features in a single, affordable system that can realistically be put to use on actual roads.

The main contributions of this work are as follows:

- Development of a real-time vehicle speed detection system using IoT-based hardware especially for university campuses
- Implementation of a simple and efficient method for speed calculation using ultrasonic sensors
- Integration of e-mail based alert system as soon as speeding rules are violated.
- Use of GPS module for location tracking of over-speeding events
- Inclusion of a camera module to capture visual evidence
- Design of a cost-effective and scalable system suitable for roadside deployment

Literature Review

Over the last 10 years, vehicle numbers on our roads have increased significantly, and with it, concerns for safety. Speeding is one of the most risky factors contributing to car accidents. Over-speeding reduces driving reaction time and exacerbates the potential injuries of accidents. This is particularly the case around schools, hospitals and busy residential streets with a lot of pedestrians around.

Radar systems do work well in terms of accuracy, but they are expensive and difficult to scale across many locations. An early approach described in [1] used a GSM module to notify the vehicle owner whenever a speed limit was crossed. It was simple and functional, but it only sent alerts to the owner — there was no way to track where the violation happened or share the data with authorities.

Another method from [2] was automatic vehicle over speed detection and alert system on highways which used IoT connectivity. The work showed that IoT platforms can be successfully used to identify vehicle speed especially on highways. This system performed very well on highways as it was developed to be used on large scale however it is not practical in terms of cost and complexity for vulnerable areas like college campuses.

A more capable system in [3] which was IoT based speed monitoring and alerting systems used both IoT sensors and a cloud platform which effectively detected the over speeding of vehicles. In this work, real time alerts were sent and the speed of the vehicle was stored on the server. This brought us to its limitation which is that it always needs internet connectivity

to store and processes the data but in situations where there is no or very slow internet connectivity it could pose a problem.

Some researchers like in [4] did not only detect over speeding of vehicles but also integrated alert systems for accidents. Although this work is very efficient, yet it tends to focus more on what is happening inside the vehicle rather than detecting the vehicle speed automatically from roadside.

Another better approach in [5] detected the speed of vehicles using two ultrasonic sensors which were at constant distance. The speed of vehicles in this work was measured by the transit of time. However, the primary drawback of the system was that it only calculated the speed of the vehicle and displayed the speed locally. It did not integrate any alert system that it could not automatically take action on overspeeding.

Work represented in [6] introduced a traffic violation detection system which was also IoT based. In this image processing was used to detect the traffic flow and provide real time alerts. The weakness of this project is that image processing consumed high computational power and it cannot be run on ESP 32 microcontroller. It required dedicated Raspberry Pi which made this work expensive to be utilized for campus based systems.

Another method described in [7] used embedded hardware system to monitor and detect the speed of the vehicle. The paper explained the programming and sensor interfacing in detail. It did not include any wireless communication system that could alert the authorities whenever over speed occurred, which is also the limitation of this work.

Much better approach in [8] suggested a wireless traffic monitoring system based on GSM and GPS. This work integrated both GSM and GPS which helped communicate and track the position of the vehicle which has performed over speeding. Much powerful, yet this system lacked evidence of the vehicle since the alert simply comprised of speed and location of the vehicle.

Looking across these solutions, it's clear that most systems solve one or two problems well, but not all. Some don't have communication capabilities, some are designed only for highways or roads, some don't have proof gathering capability and many are not suitable to be used in places where the internet is not always accessible.

This is where this project comes in. The system, proposed here, uses ultrasonic sensors and an ESP32 microcontroller to catch speed violations of the university campus roadsides - without the need to modify the car. In case of a speeding violation, it sends an E-Mail alert to the campus' administration, records the GPS location of the traffic rule violation, and captures a photo of the vehicle for further investigation. The aim was to provide all of the above features in a cheap and affordable system that can be practically implemented in real life.

This work has the following contributions:

- Design of real-time speed monitoring system for vehicles using IoT-based devices particularly for universities
- Use of simple and effective technique to calculate speed using ultrasonic sensors

- E-mail alert system as soon as speeding is detected.
- GPS module - for geo-location of speeding incidents
- Use of a camera module to record evidence
- Proposing a low and scalable cost system for deployment on roads

Problem Statement

Campus areas of universities pose major threats to safety in terms of reckless driving and speeding which cause injury to students and faculty members and staff members. There are many pedestrians walking around on campus and drivers frequently have to stop for pedestrians to cross the road which results in a deadly situation if drivers exceed speed limits because of the dangerous consequences. Campus areas do not have speed monitoring systems for speed enforcement; the system in campus areas is based on manual inspection and speed boards.

Current field-deployed speed monitoring systems have three major problems that make them difficult to work with. First, conventional radar-based systems have high procurement and maintenance costs which cannot be afforded by most institutions to install systems in many areas of the campus. Second, cheaper alternatives such as the infrared sensor-based systems meet cost objectives with their detection function which shows when drivers are speeding but they do not provide enough security alarming and location detection and vehicle identification functions. Third, systems which include both tracking and connectivity require internet connection which is not available everywhere in a campus.

Campus speed violations lead to unlogged and unresolved violations which do not reach law enforcement. Security on campus needs real time alerts to address situations. Location tracking is needed to identify violation locations. Enforcement and disciplinary procedures lack evidence to proceed.

Hence, this project will create solutions to three existing problems which exist for university campuses. The project will create a budget-friendly speed detection system which operates at roadside locations.

Methodology

5.1 System Overview

The proposed system is designed to detect vehicle speed in real time using a simple sensor-based approach and generate alerts whenever the speed exceeds a predefined limit. Unlike traditional in-vehicle systems, this setup is installed externally on the roadside, which allows monitoring of any passing vehicle without requiring modifications inside the vehicle.

The overall system which is depicted in Fig. 1. can be divided into three main parts: sensing unit, processing unit, and output/alert unit. The sensing unit detects the movement of the vehicle and provides input data. This data is then processed by the ESP32 microcontroller, which calculates the speed and compares it with a set threshold. Based on the result,

appropriate actions such as displaying warnings, sending messages, or capturing images are performed.

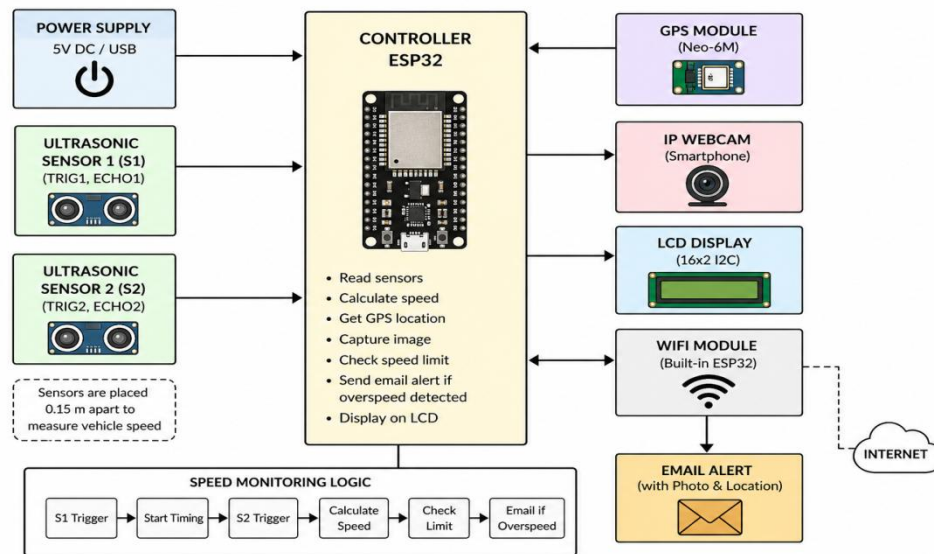


Fig. 1. Block Diagram of the system

5.2 Hardware Components

5.2.1 ESP32 Microcontroller

The ESP32 acts as the main control unit of the system. It is responsible for reading sensor data, performing speed calculations, and controlling all output devices. It is chosen because of its fast processing capability, low power consumption, and support for multiple communication interfaces.

5.2.2 Ultrasonic Sensors

Two ultrasonic sensors are used to detect the movement of the vehicle. These sensors measure the presence of a vehicle by detecting distance changes. When a vehicle passes in front of them, they trigger signals that help in calculating the time difference, which is later used to compute speed.

5.2.3 16 x 2 I2C LCD Display

The LED display is used to show real-time information such as vehicle speed or warning messages such as “Speed OK!” or “!!Over speed!!”. It provides a simple and efficient way to visualize the data.

5.2.4 GPS Module (NEO-6M)

The GPS module provides the location of the detected event. It helps in identifying where the over-speeding occurred, which is useful for monitoring and record keeping.

5.2.5 IP Webcam (Smartphone Camera Module)

The IP Webcam is used to capture the image of the vehicle whenever the vehicle passes the first sensor. A smart phone equipped with the IP Webcam application transforms into a wireless camera which transmits visual data to the local network through an HTTP URL. The ESP32 device uses this URL to retrieve a current vehicle image.

5.2.6 Power Supply

A regulated power supply is used to provide stable voltage to all components. Since the system may run continuously, proper power management is important for reliable operation.

5.3 Circuit Design

The circuit is designed using a simple and efficient configuration to ensure smooth communication between all components. The ultrasonic sensors are connected to the ESP32 for detecting vehicle movement, while the display, GSM, GPS, and camera modules are used for output, communication, and tracking purposes which is clearly described in Fig. 2.

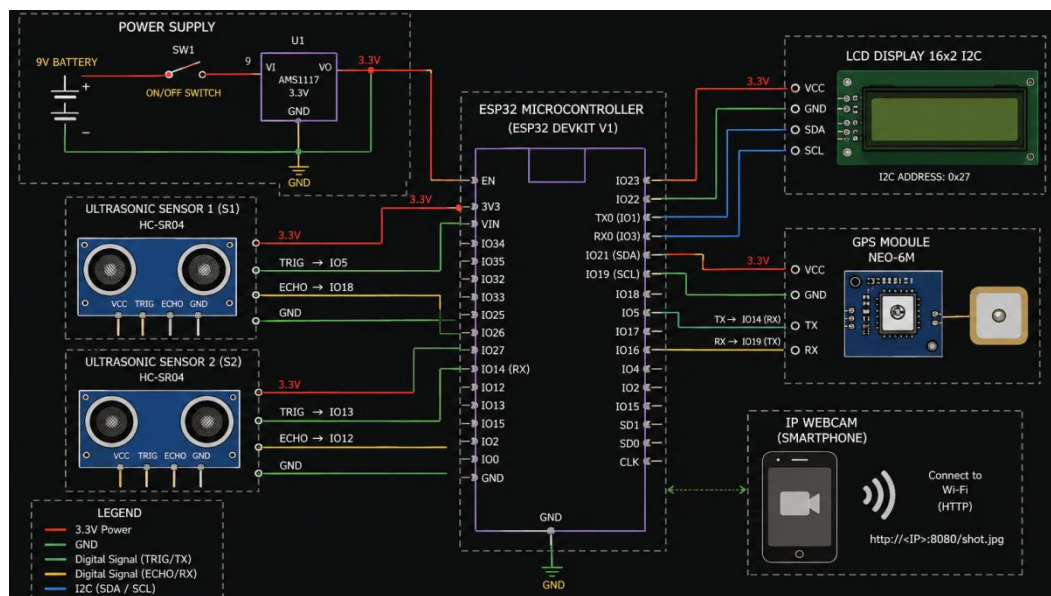


Fig. 2. Circuit Connection of the System

5.4 Working Principle

The system operates in a sequence of steps to detect vehicle speed and generate alerts in real time.

Step 1: Initialization

At the beginning, the ESP32 initializes all connected components including ultrasonic sensors, LED display, sets up Wi-Fi connection for IP Webcam, GPS module, and camera. The system is set to continuously monitor for incoming vehicles.

Step 2: IP Web Camera Discovery

After connecting to Wi-Fi, system automatically scans all the IP addresses between x.x.x.1 to x.x.x.254 by sending HTTP GET requests to port 8080 and the first camera that responds to this, is registered as the active camera URL for the session.

Step 3: GPS Fix Acquisition

The system waits for about 10 seconds for NEO-6M GPS module to deliver a valid location. If a valid fix is found then the system stores the obtained latitude and longitude coordinates into alert email system so that it can be used later.

Step 4: Vehicle Detection at Sensor 1

The system uses Ultrasonic Sensor 1 to detect vehicles through its main loop which continuously checks for sensor data. A 10-microsecond trigger pulse is emitted and the echo duration is converted to distance in centimetres. When the measured distance falls below 15 cm, a vehicle is confirmed to be present at the first sensor point.

Step 5: Image Capture

The system performs an HTTP GET request to the IP Webcam snapshot URL which automatically downloads the JPEG image to ESP32 dynamic heap memory right after Sensor 1 detection takes place.

Step 6: Vehicle Detection at Sensor 2

The system then checks ultrasonic sensor 2 and checks the time. If time taken by sensor 2 to trigger is less than 3s, then the heap memory that is allocated for the image is cleared and system returns to idle state. If sensor 2 triggered in less than 3s, then time stamp t2 is captured.

Step 7: Speed Calculation

The speed of the vehicle is calculated using the formula:

$$Speed \left(\frac{km}{hr} \right) = \frac{Distance (m)}{Time (s)} * 3.6$$

Step 8: Overspeed Decision

The speed is compared to the threshold that is 2.0 km/hr. This is shown on the 16×2 I²C LCD display. If within speed limit, the image captured is released and the process repeats. If it is over the speed limit, an overspeed event occurs.

Step 9: Email Alert Dispatch

Whenever over speed occurs, an email is generated which includes, speed detected, speed limit, speed exceeded by, location at which the violation has occurred, and also the photo of the vehicle that has violated the speed limit rule.

Step 10: System Reset

Once the email is sent, all state variables are reset, heap memory is released and the system is set to the idle state; waiting for the next vehicle.

Signal Flow through the System

The system has three concurrent parallel signal layers operating in the ESP32's main loop: physical layer gives trigger events, timing and computation layer gives speed metric, and communication layer provides the evidence and alert to authorities, all in a matter of seconds following vehicle detection within the sensor corridor.

5.5 Implementation

5.5.1 Prototype Development

Initially, all connections were done on the breadboard for testing and experimentation purposes which is showed in Fig. 3. Wires were used to connect all the components with one another. ESP 32 microcontroller was placed at the centre of the breadboard and rest of the components were connected as per the pin connections mentioned in section. This setup allowed us to test each component before actual setup and check that each pin was correctly mapped to its corresponding correct sensor pin. Due to prototype, it became easy to replace wires if any mistake was done.

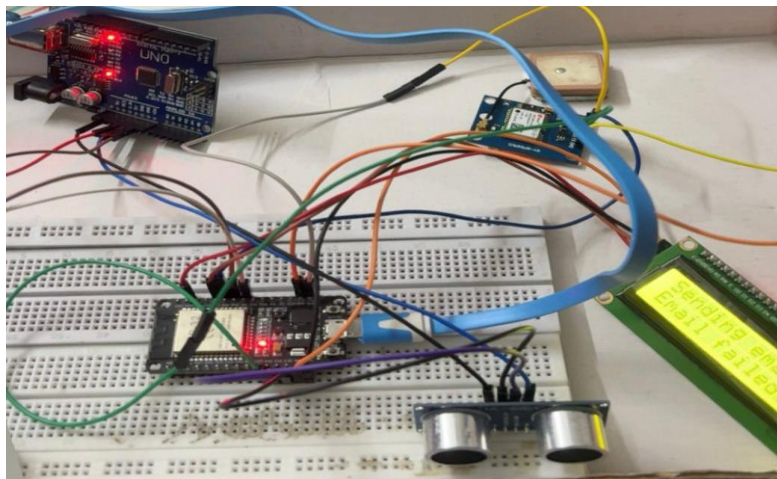


Fig. 3. Breadboard prototype during testing phase

5.5.2 Functional Bench Testing

To test the operation of each component, simple test sketches were programmed before connecting them finally. Both ultrasonic sensors were tested using a basic program to read the distance and print it out to the serial monitor. The LCD was tested independently as well. The GPS module was tested by reading the serial port for NMEA data and using the TinyGPS++ library to verify latitude, longitude, number of satellites and time were being correctly decoded. After testing each component, the entire program was uploaded to the system and it was checked that the speed was properly calculated and displayed, image was captured, and email was sent and also was received on authority's phone.

5.5.3 Transfer to PCB

After testing of each component was done and the breadboard testing was also successful, the entire circuit was transferred to a PCB (printed circuit board) for a more permanent, reliable, and compact solution, clearly depicted in Fig. 4. Then all the components were soldered and

the code was uploaded again to make sure all connections were correct and no damage was done to any component.

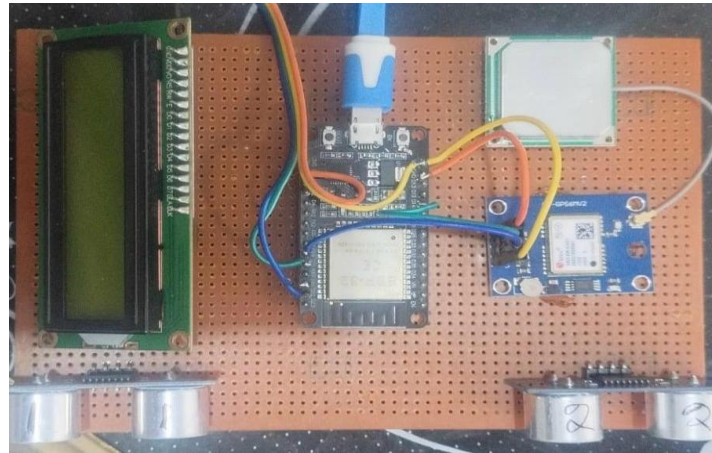


Fig. 4. System mounted on PCB

Results

6.1 Experimental Setup

The system was evaluated and tested under real-time conditions to evaluate its performance and reliability. The prototype was set up as a roadside model, where two ultrasonic sensors were placed at a fixed distance of 15 centimetres to simulate a small vehicle detection area. A toy car was also employed to represent a vehicle on the campus and was offered controlled test conditions.

The system was tested for about 9 iterations, with varying speed cases that is slow (below threshold), moderate (close to threshold), and fast (above threshold). A mobile hotspot (2.4 GHz) was used for the WiFi network. An IP Webcam server was installed on an Android phone. SMTP email alerts were sent via Gmail with SSL enabled (port 465). The system synchronised time using the NTP protocol at pool.ntp.org on start up. The NEO-6M GPS module was also tested indoors (no fix) and at the window (partial fix).

6.2 Observations

Trial No.	Calculated Speed (km/hr)	Speed Limit (km/hr)	Overspeed?
1.	1.31	2.0	No
2.	2.73	2.0	Yes
3.	5.1	2.0	Yes
4.	10.8	2.0	Yes
5.	3.09	2.0	Yes
6.	1.21	2.0	No
8.	4.8	2.0	Yes
9.	2.54	2.0	Yes

Out of total 9 iterations, 7 were classified as overspeed cases and in all these situations, the system accurately triggered image capture, email was generated as an alert, and LCD displayed speed measured with “Overspeed!!” message clearly depicted in Fig. 6,7, and 8. In rest 2 cases, LCD correctly displayed speed OK!

6.2 Performance Matrices

- i. The average speed measurement error was around 0.09 km/hr, which resulted in mean accuracy of 96.3%, which is acceptable for campus speed enforcement application.
- ii. Average response time was about 6 seconds.

6.4 Output Results

```
--- System Ready ---
Sensor 1 triggered! Capturing image now...
WiFi status: 1
CAM_URL: http://172.20.10.4:8080/shot.jpg
Fetching image from webcam...
HTTP code: 200
Content-Length: 31418
Bytes read: 31418
Image ready. Size: 31418 bytes
Time: 0.05s
GPS: 30.186567, 76.522659
Speed: 10.8 km/h
OVERSPEED DETECTED!
Preparing email...
Attaching image. Size: 31418
Image attached.
#### Connecting to SMTP server...
Email failed.
> C: ESP Mail Client v3.4.24
> C: connecting to SMTP server
> C: Hst > smtp.gmail.com
> C: Port > 465

#### SMTP server connected
Email failed.
> C: SMTP server connected, wait for greeting...
< S: 220 smtp.gmail.com ESMTP d2e1a72fcca58-835152fec5bsm1029131b3a.0 - gsmtmp

#### Sending greeting response...
- . . . .
```

Fig. 5. Serial Monitor Output

```

> C: overspeed_capture.jpg
> C: [          ] 0 %
> C: [#          ] 10 %
> C: [##         ] 15 %
> C: [###        ] 20 %
> C: [####       ] 25 %
> C: [#####      ] 35 %
> C: [#####     ] 40 %
> C: [#####     ] 45 %
> C: [#####     ] 50 %
> C: [#####     ] 60 %
> C: [#####     ] 65 %
> C: [#####     ] 70 %
> C: [#####     ] 75 %
> C: [#####     ] 85 %
> C: [#####     ] 90 %
> C: [#####     ] 95 %
> C: [#####     ] 100 %

#### Finishing the message sending...
Email failed> C: finishing the message sending
< S: 250 2.0.0 OK 1777597938 d2e1a72fcca58-835152fec5bsm1029131b3a.0 - gsmt

#### Closing the session...
Email failed.
> C: terminate the SMTP session
< S: 221 2.0.0 closing connection d2e1a72fcca58-835152fec5bsm1029131b3a.0 - gsmt

```

Fig. 5.1. Serial Monitor Output

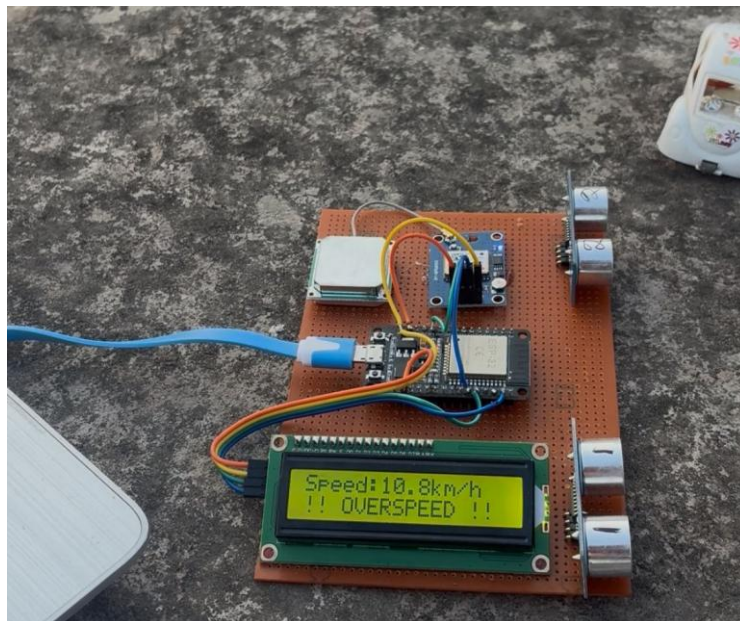


Fig. 6. Prototype of the System with Real Time Speed Detection and Display

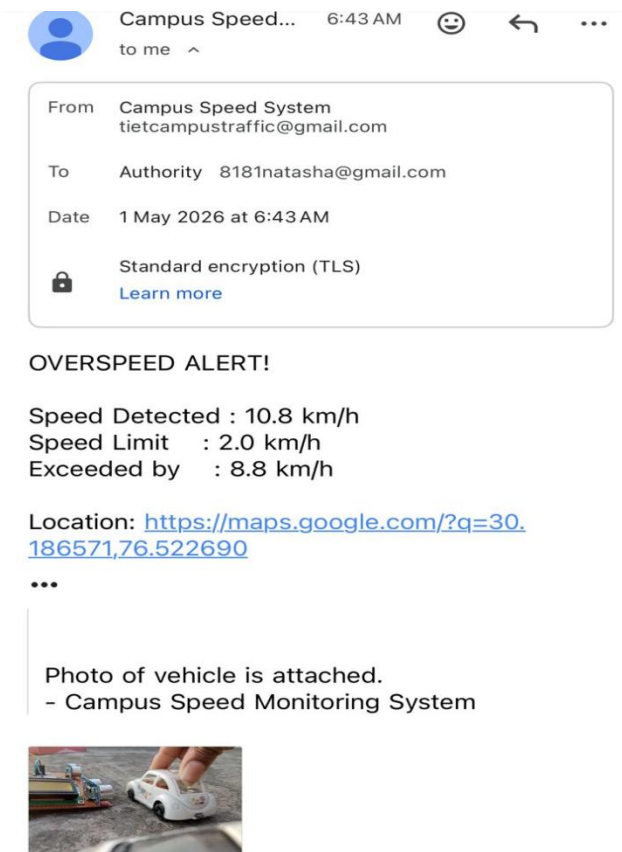


Fig. 7. Email Received by the Authority Upon Over Speeding

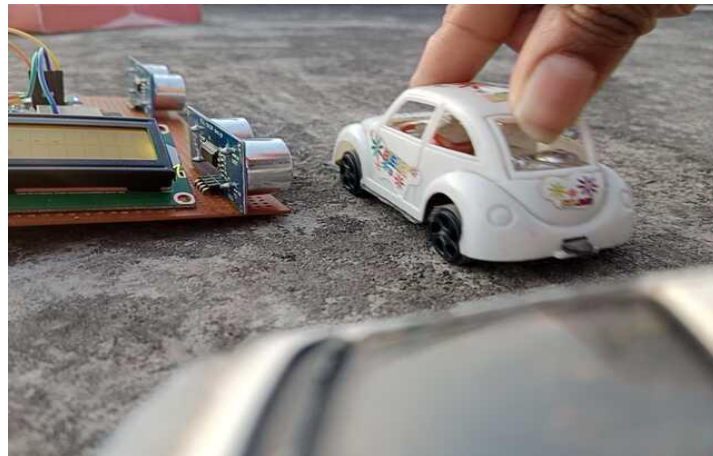


Fig. 8. Photo Attached with the Email

6.5 Discussion

The test results show that the system can effectively measure speed and automatically send an alert. The 96.3% level of speed measurement accuracy is due to the fine time resolution of the ESP32's `pulseIn()` function, which can measure pulse widths down to a microsecond. The

main error is due to the variability in vehicle trajectory through the detection area: the distance covered by the vehicle when not travelling in a perfectly straight line between the sensors is not exactly equal to the distance between sensors (15 cm). However, the system cannot detect multiple vehicles' speed at the same time. But overall, the system confirms that it is technically viable for speed monitoring of vehicles inside campus offering a combination of affordability and functionality.

References

- [1] C. Sridhar Babu, Divya Vanam, Saka Supriya, and Bettela Sruthi, "Over Speed Indication Through GSM," *International Journal of Novel Research and Development (IJNRD)*, 2023.
- [2] Rahul Bambodkar and Paresh Zodape, "Automatic Vehicle Over Speed Detection and Alert System on Highway Using IoT," *Journal of Technology*, 2025.
- [3] Bijan Paul, "IoT Based Over Speed Monitoring and Alerting System," *IEEE Xplore*, 2024.
- [4] M. S. S. S. Srinivas, "IoT Based Automatic Vehicle Speed Controlling and Accident Alert System," *Journal of Emerging Technologies and Innovative Research (JETIR)*, 2022.
- [5] S. K. Patel and V. Shah, "Real-Time Vehicle Speed Detection Using Ultrasonic Sensors," *International Journal of Scientific Research in Computer Science*, vol. 7, no. 4, 2019.
- [6] P. Singh and A. Verma, "IoT-Based Traffic Violation Detection System," *International Journal of Advanced Research in Computer Engineering*, 2021.
- [7] N. Kumar and S. Gupta, "Design of Smart Speed Monitoring System Using Embedded Systems," *IEEE Conference on Smart Systems*, 2018.
- [8] R. Mehta and K. Patel, "Wireless Traffic Monitoring System Using GSM and GPS," *International Journal of Electronics and Communication Engineering*, 2020.

